

# Class 12 Physics - Electric Charges and Fields

## *Practice Questions with Solutions*

**Q1: Which of the following organisms primarily reproduces by binary fission?**

- A) Yeast
- B) Amoeba
- C) Hydra
- D) Penicillium

**Answer:** B

**Solution:** **\*\*Amoeba\*\*** is a unicellular organism that reproduces asexually by binary fission, where the parent cell divides into two daughter cells.

**Q2: What is the period from birth to the natural death of an organism called?**

- A) Reproductive phase
- B) Juvenile phase
- C) Life span
- D) Senescent phase

**Answer:** C

**Solution:** The **\*\*life span\*\*** is defined as the period from birth to the natural death of an organism.

**Q3: In which of the following organisms are spores commonly found as a mode of asexual reproduction?**

- A) Earthworm

- B) Fungi
- C) Mammals
- D) Flowering plants

**Answer:** B

**Solution:** **\*\*Fungi\*\*** commonly reproduce asexually through the formation of various types of spores, such as zoospores, conidia, etc.

**Q4: Offspring formed by asexual reproduction are called:**

- A) Zygotes
- B) Gametes
- C) Clones
- D) Embryos

**Answer:** C

**Solution:** Offspring produced by asexual reproduction are genetically and morphologically identical to the parent and are thus referred to as **\*\*clones\*\***.

**Q5: Identify the plant that reproduces vegetatively by adventitious buds present on the leaf margin.**

- A) Potato
- B) Bryophyllum
- C) Ginger
- D) Onion

**Answer:** B

**Solution:** **\*\*Bryophyllum\*\*** reproduces vegetatively through adventitious buds that arise from the notches present on the margins of its leaves.

**Q6: The fusion of male and female gametes is called:**

- A) Gametogenesis
- B) Fertilization
- C) Embryogenesis
- D) Parthenogenesis

**Answer:** B

**Solution:** The process of fusion of male and female gametes is known as **fertilization**, leading to the formation of a zygote.

**Q7: Which of the following is a hermaphrodite animal?**

- A) Cockroach
- B) Human
- C) Earthworm
- D) Frog



**Answer:** C

**Solution:** **Earthworm** is a hermaphrodite (bisexual) animal, meaning a single individual possesses both male and female reproductive organs.

**Q8: What is the ploidy level of the zygote?**

- A) Haploid ( $n$ )
- B) Diploid ( $2n$ )
- C) Triploid ( $3n$ )
- D) Tetraploid ( $4n$ )

**Answer:** B

**Solution:** The zygote is formed by the fusion of two haploid gametes ( $n+n$ ), resulting in a **diploid ( $2n$ )** cell.

### Q9: Asexual reproduction is common among:

- A) Single-celled organisms
- B) Plants with simple organization
- C) Animals with simple organization
- D) All of the above

**Answer:** D

**Solution:** Asexual reproduction is prevalent in **single-celled organisms**, and also in **plants and animals with relatively simple organization**, making "All of the above" the correct answer.

### Q10: Which of the following structures is NOT involved in asexual reproduction in plants?

- A) Rhizome
- B) Bulb
- C) Pollen grain
- D) Tuber



**Answer:** C

**Solution:** **Pollen grain** is involved in sexual reproduction as it carries the male gametes. Rhizome, bulb, and tuber are all vegetative propagules used in asexual reproduction.

### Q11: Define reproduction. Why is it essential for organisms?

**Solution:** **Reproduction** is a biological process by which organisms produce offspring identical or similar to themselves. It is essential for the **continuation of species**, ensuring the survival of life on Earth and maintaining the population of a species across generations.

## Q12: Differentiate between asexual and sexual reproduction.

**Solution:** **Asexual reproduction** involves a single parent producing offspring that are genetically identical to the parent, with no fusion of gametes. **Sexual reproduction** involves two parents (usually) producing offspring that are genetically different from the parents, through the formation and fusion of gametes.

## Q13: What are zoospores? Give an example of an organism that reproduces by zoospores.

**Solution:** **Zoospores** are microscopic, motile (flagellated) asexual spores. They are commonly found in simple organisms like **Chlamydomonas** (an alga) for reproduction.

## Q14: Explain the term 'clone' in the context of reproduction.



**Solution:** A **clone** refers to a group of individuals that are morphologically and genetically identical to each other and to their parent. Clones are typically produced through asexual reproduction.

## Q15: Name any two vegetative propagules found in plants and mention the plant they are found in.

**Solution:** Two vegetative propagules are:

## Q16: What is the significance of meiosis during sexual reproduction?

**Solution:** **Meiosis** is crucial during sexual reproduction as it reduces the chromosome number by half in gametes, ensuring that the zygote formed after fertilization has the correct

diploid number of chromosomes specific to the species. It also introduces genetic variation through crossing over.

### Q17: Define external fertilization and give one example.

**Solution:** **External fertilization** is the process where the fusion of male and female gametes occurs outside the body of the organism, typically in an aquatic environment. An example is **frog** or most **fish**.

### Q18: What is parthenogenesis? Give an example.

**Solution:** **Parthenogenesis** is a form of asexual reproduction where a new organism develops from an unfertilized egg. An example is **rotifers** or **honeybees** (drones develop from unfertilized eggs).

### Q19: Briefly describe the juvenile phase in the life cycle of an organism.



**Solution:** The **juvenile phase** (also known as the vegetative phase in plants) is the period of growth and maturity in an organism's life before it becomes reproductively active. During this phase, the organism is not capable of reproduction.

### Q20: How do fungi reproduce asexually?

**Solution:** Fungi primarily reproduce asexually through the formation of various types of **spores**, such as conidia (e.g., *Penicillium*) or zoospores (e.g., some aquatic fungi). They can also reproduce by budding or fragmentation.

**Q21: Describe different types of asexual reproduction commonly observed in organisms, with suitable examples for each.**

**Solution:** Asexual reproduction is a mode of reproduction that does not involve the fusion of gametes and results in offspring that are genetically identical to the single parent. Here are common types with examples:

**Q22: Explain the events in sexual reproduction, detailing pre-fertilization, fertilization, and post-fertilization events.**

**Solution:** Sexual reproduction is a complex process involving the formation and fusion of gametes. It can be divided into three main sequential events:

**Q23: Describe the different phases of the life cycle of an organism.**

**Solution:** The life cycle of an organism generally involves three distinct phases:

**Q24: With the help of diagrams, explain budding in Yeast and Hydra.**

**Solution:**

**Q25: Differentiate between internal and external fertilization, providing advantages and disadvantages of each.**

**Solution:**

**Q26: Discuss the different types of sexual reproduction in terms of gamete morphology and behavior.**

**Solution:** Sexual reproduction involves the fusion of gametes, and based on the morphology and behavior of these gametes, we can categorize sexual reproduction into a few types:

**Q27: Explain the significance of post-fertilization changes in a flowering plant.**

**Solution:** Post-fertilization changes in flowering plants are crucial for the development of the seed and fruit, which are vital for the survival and dispersal of the next generation. These changes include:

**Q28: Describe various methods of vegetative propagation in plants.**



**Solution:** Vegetative propagation is a form of asexual reproduction in plants where new plants are produced from vegetative parts of the parent plant, such as roots, stems, or leaves, without the involvement of seeds or spores. This results in genetically identical offspring (clones). Various natural and artificial methods exist:

**Q29: What is the significance of sexual reproduction despite its complexities?**

**Solution:** Despite being a more complex and time-consuming process compared to asexual reproduction, sexual reproduction holds immense evolutionary and biological significance:

**Q30: Describe the process of embryogenesis in an animal, highlighting the key stages.**

**Solution:** **\*\*Embryogenesis\*\*** is the complex process of development of an embryo from the zygote following fertilization. It involves a series of coordinated cell divisions, growth,

and differentiation, ultimately leading to the formation of a complete organism. While the details vary across animal species, the key stages are generally:

